

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (CL) Examination****March-April 2018****CL409/CL409.01 Construction Management****Date: 30.03.2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70*****Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. No graph paper will be provided for Barchart.

SECTION - I**Q.1 Answer the following questions.****[07]**

- i. Enlist important reasons due to which construction projects are generally failed or delayed.
- ii. Total float is the duration by which an activity can be delayed without delaying any succeeding activity. True or False?
- iii. Enlist functions of construction management.
- iv. Which kind of scheduling technique will be used for a highway construction project?
- v. Construction projects are gradually becoming multidisciplinary with multiple objectives. True or False?
- vi. What do you understand by critical path?
- vii. Based on the following statements select the appropriate option.
 - 1) There can be only one critical path in a project network
 - 2) It is necessary to introduce a dummy activity in AON network to ensure logic
 - 3) The critical path is the longest path in the network

a) 1– True; 2&3- False

b) 2– True; 1&3- False

c) 1,2– True; 3- False

d) 3- True; 1,2- False

Q.2 (a) Describe various stages of construction management with illustration.**[05]****Q.2 (b) Enlist and explain various methods of scheduling in Construction Project Management and compare based on its suitability.****[05]****OR****Q.2 (b) Explain functional organisation with its suitability, advantages and disadvantages.****[05]****Q.2 (c) Draw schematic diagram of Project feasibility study.****[04]****OR****Q.2 (c) Enlist and explain the types of analysis carried out for feasibility study.****[04]**

- Q.3 (a)** Using the following data given in following table, determine total duration of project and critical path of project. **[04]**

Activity	Duration (Days)	Predecessor
A	2	-----
B	4	A
C	3	A
D	6	B, C
E	7	B
F	3	D, E

- Q.3 (b)** Attempt any two questions. **[10]**

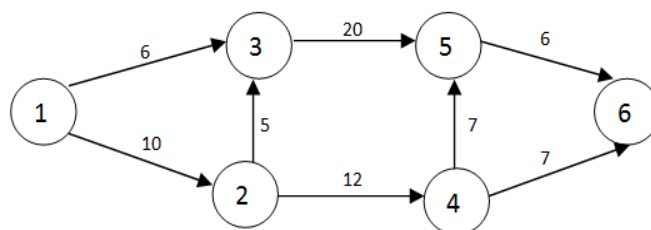
- i. The following table lists the activities and their respective duration for a project. Find out critical path, total float, free float, independent float, interfering float for listed activities.

Activity	Duration (Days)	Activity	Duration (Days)
1-2	6	3-6	4
1-3	7	4-5	9
1-4	8	5-6	5
2-5	7	5-7	6
2-7	8	6-7	6
3-4	4	7-8	9

- ii. Below given table shows activities and its related costs for a small construction project. Overhead costs of the project is Rs. 2000/- per week. Find the optimum duration and the cost associated with it.

Activity	Normal duration (weeks)	Normal cost (Rs.)	Crash duration (weeks)	Crash cost (Rs.)
1 - 2	4	4000	2	12000
2 - 3	5	3000	2	7500
2 - 4	7	3600	5	6000
3 - 4	4	5000	2	10000

- iii. Update the network given below. Compare the critical paths for the original network and updated network. Also calculate the updated duration of project.



After 14 days working on the project, the following conditions exist:

- Activity 1-2 and 1-3 are completed as an originally planned.

- Activities 2-3 is in process and will need 1 more day for completion.
- Activities 2-4 is in process and will need 16 more days for completion.
- Activity 4-5 appears to present some problem and now its estimated time of completion is 10 days.
- Activity 4-6 can be completed in 5 days instead of originally planned 7 days.
- All other activities duration remain unchanged.

SECTION - II

Q - 4 Answer the following questions.

- i. What is work breakdown structure of construction project? **[01]**
- ii. Explain with Sketch: Parallel Activity, Serial Activity **[02]**
- iii. Define the terms: Optimistic Time, Pessimistic Time. **[02]**
- iv. With the schematic diagram explain head event and tail event in CPM. **[02]**

Q.5 (a) The breakdown of certain project into various activities is as under. Prepare bar chart for the project. Also find out the total time of completion of the project. **[04]**

Activity	A	B	C	D	E	F	G
Duration(weeks)	2	4	6	3	1	2	5

- Activity B & C can be performed concurrently and they both must follow activity A.
- Activity B must precede activity D.
- Activity E cannot begin until both activity B & C are complete.
- Activity F can be started only after activities D & E are complete.
- Activity G is the last activity which can be started only after completion of activity E

OR

Q.5 (a) A small project consists of six activities designated from A to F, with following relationships. Draw an AON network and number the events. **[04]**

- A is the first job to be performed.
- B and C can be done concurrently and must follow A.
- B must precede D.
- G follows C and D.
- E must succeed B.
- The last operation F is dependent on completion of both E and G.

Q.5 (b) Briefly explain Payment of Wages Act. **[03]**

Q.5 (c) Using the following data given in table, determine critical path and probability of completion of the project within 35 days. (Refer normal distribution function table given **[07]**

on page no. 4)

Activity	t _o (days)	t _m (days)	t _p (days)
1-2	6	9	18
1-3	5	8	17
2-4	4	7	22
3-4	4	7	16
4-5	4	10	22
2-5	4	7	10
3-5	2	5	8

Q.6 (a) Attempt any two questions.**[10]**

- What is a tender? What are the details included in tender notice?
- Enlist various types of contract used in civil engineering projects and explain any two types of contract in detail.
- Explain the mode of submission of tender.

Q.6 (b) Prepare work breakdown structure for residential construction project.**[04]****Reference Table: Normal distribution function (Q.5(C))**

Normal Deviate –	Probability (%)	Normal Deviate +	Probability (%)
0	50.0	0	50.0
– 0.1	46.0	+ 0.1	54.0
– 0.2	42.1	+ 0.2	57.9
– 0.3	38.2	+ 0.3	61.8
– 0.4	34.5	+ 0.4	65.5
– 0.5	30.8	+ 0.5	69.2
– 0.6	27.4	+ 0.6	72.6
– 0.7	24.2	+ 0.7	75.5
– 0.8	21.2	+ 0.8	78.8
– 0.9	18.4	+ 0.9	81.6
– 1.0	15.9	+ 1.0	84.1
– 1.1	13.6	+ 1.1	86.4
– 1.2	11.5	+ 1.2	88.5
– 1.3	9.7	+ 1.3	90.3
– 1.4	8.1	+ 1.4	91.3
– 1.5	6.7	+ 1.5	93.3
– 1.6	5.5	+ 1.6	94.5
– 1.7	4.5	+ 1.7	95.5
– 1.8	3.6	+ 1.8	96.4
– 1.9	2.9	+ 1.9	97.1
– 2.0	2.3	+ 2.0	97.7
– 2.1	1.8	+ 2.1	98.2
– 2.2	1.4	+ 2.2	98.6
– 2.3	1.1	+ 2.3	98.9
– 2.4	0.8	+ 2.4	99.2
– 2.5	0.6	+ 2.5	99.4
– 2.6	0.5	+ 2.6	99.5
– 2.7	0.3	+ 2.7	99.7
– 2.8	0.3	+ 2.8	99.7
– 2.9	0.2	+ 2.9	99.8
– 3.0	0.1	+ 3.0	99.9
